MOHAMMAD MOUSTAFA

MSc 1 ARMAC Student | Automation, Robotics & Computer Vision

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PROFILE

MSc 1 ARMAC student (Automation, Robotics, Mechatronics and Control) at Université Marie et Louis Pasteur, Besançon, France. Mechatronics engineer by training (Bachelor of Engineering, Rafik Hariri University, Lebanon), with hands-on experience in industrial automation (Siemens PLCs, HMI, variable frequency drives), robotics (Arduino, Raspberry Pi, ROS 2) and computer vision (OpenCV, object detection, face recognition). Additional background in industrial maintenance and in training/supervising students on technical projects.

PROFESSIONAL EXPERIENCE

Robotics Instructor (Part-time) **Jan. 2025 – Aug. 2026**

AL-JANA (NGO), Sebline, Lebanon

- Delivered hands-on robotics training for university-level students using Arduino, Raspberry Pi and sensors/actuators.
- Introduced the working principle, types and applications of each component before its use in a project.
- Supervised computer vision projects using a Raspberry Pi camera (object detection, face recognition) with OpenCV.
- Provided technical supervision to students throughout the implementation of their robotics projects.

Industrial Automation- Internship **July 2026**

Ciment de Sibline S.A.L, Sibline, Lebanon

- Immersion in the control and automation room of a cement plant: observation of PLCs, HMIs and large-scale production processes.
- Exposure to robotic packaging systems (pick-and-place) and industrial equipment in operation.

Automation & Control Engineer – Internship **April – June 2026**

Matrix EA Academy, Khaldeh, Lebanon

- Reverse-engineering of industrial automation systems: PLCs, HMIs, variable frequency drives (VFD), sensors, hydraulic and pneumatic systems.
- Prepared mechanical, hydraulic, electric, control, and pneumatic technical documentation using EPLAN, Fluid Sim, Solidworks, WinccFlexible, and TIA Portal.

Assistant Engineer – Internship **July – Sept. 2024**

UNRWA–KFW, Beirut, Lebanon

- Contributed to the maintenance of generators, vehicles and solar installations.
- Gained practical experience in photovoltaic (solar) system installation.

Industrial Maintenance Technician (Consolidated Roles) **2022 – 2024**

SuperPlast, Jadra, Lebanon

- Executed maintenance operations on plastic injection molding production lines, including mold changes, fault diagnostics, and downtime reduction.

EDUCATION

MSc 1 ARMAC – Automation, Robotics, Mechatronics and Control **2026 – 2028**

Université Marie et Louis Pasteur, Besançon, France

Bachelor of Engineering (B.Eng.) in Mechatronics Engineering- ABET Accredited **2022 – 2026**

Rafik Hariri University, Lebanon (BAC+5 year foreign engineering degree)

ACADEMIC & PERSONAL PROJECTS

- **Industrial Chemical Blending PLC control system:** Siemens S7-1200, TIA Portal, Ladder Logic, HMI, analog and digital inputs.
- **Recognition System (OpenCV / Python):** Implemented real-time image processing algorithms for object tracking, filtering, and detection via embedded video streams.

- **Self-Balancing Electric Unicycle (PID Control):** Complete mechatronic design involving IMU sensor fusion, BLDC motor control, and closed-loop PID tuning.
- **Software Architecture & Smart-Home Simulation (ROS 2**): Developed a complete software architecture under ROS 2 for autonomous equipment management and robotic navigation.
- **Quadcopter:** design and assembly of a 4-rotor flying platform.
- **Automated coffee machine:** Arduino Mega, I2C LCD display, keypad, DS18B20 temperature sensor, pump, servo motor.
- Building Management System.
- **Automated page-flipping mechanism:** Suction motor, voice/foot-commanded control, assistive technology for paralyzed individuals.
- **Autonomous Firefighting Robot:** Designed a mobile robot equipped with fire detection sensors, closed-loop feedback control, and automated intervention capabilities.

TECHNICAL SKILLS

Industrial automation: Siemens PLCs (TIA Portal), Ladder Logic, WinCC Flexible, HMI, variable frequency drives (VFD), EPLAN, AutoCAD Electrical, Fluid SIM.

Robotics & embedded systems: Arduino, Raspberry Pi, ESP32, ROS 2, sensors, actuators

Computer vision & programming: Python, C++, OpenCV, MATLAB/Simulink

CAD / mechanical design: SolidWorks, AutoCAD

Energy: Photovoltaic system installation (hands-on experience)

Office tools: Microsoft Office suite

LANGUAGES

French: B1- Currently Learning B2

English: C1- fluent

Arabic: native language

VOLUNTEERING

Event Coordinator & Workshop Facilitator **2022 – 2025**

AL-JANA (NGO), Lebanon

- Created educational content, including lesson plans, assessments, and learning resources.
- Organized logistics and coordinated awareness workshops.
- Handled data entry, report writing and documentation management.